

CHE 321 - Exam 1 - Sample 1

exam1__1.m

```
1 function h = exam1(w,y)
2 X= y(1);
3 p= y(2);
4
5 % parameter
6 %since Ct0 given and only A then it must be Ca0
7 Ca0= 2.0;
8 Fa0= 10.0 ;
9 k=10.0 ;
10 alpha=.07;
11 eps= 3.0 ;
12
13
14 Ca=((Ca0*(1-X))*(1+eps*X))*p;
15 Cb=(Ca0*(1+X)/(1+eps*X))*p;
16 Cc=((Ca0*(3+3*X))/(1+eps*X))*p;
17
18
19 % rate eq
20 ra= -k*Ca - k*Cb*(Cc^3);
21
22 %eqns
23 h1= -ra/Fa0 ;
24 h2= (-alpha/2*p)*(1+eps*X);
25
26 h= [h1, h2]';
27
28 end
```

exam1_sim____1.m

```
1  clc;
2  clear all
3
4  % weight of particle we care about
5  wspan = [ 0 , 10];
6  % initial conditions for w and y
7  y0= [ 0 1];
8  % runs differential solver Rungekutta
9  [w,y] = ode45(@exam1,wspan,y0) ;
10
11 % Plot the results
12 figure(1);
13 plot(w, y(:, 1));
14
15 xlabel('Weight');
16 ylabel('conversion');
17 figure(2);
18 plot(w, y(:, 2));
19 xlabel('Weight');
20 ylabel('p');
21 title('Solution of ODE');
22
23 %trying to relate conversion to Dp
```

exam1_Q2____1.m

```
1  %plot conversion versus catalyste diameter ranging from .1mm to 10mm
2  %particle diameter means more void space
3  %more void space means more flow but also less surface area
4  % anytime you increase delta p with cause conversion loss
5  %having more void space means delta p decreases
6  %if delta p is decreasing, then conversion must be increasing
7
8  %given; Alpha is inversely proportional to Dp^2
9
10 %alpha= .07;
11 %p=1;
12 %esp=3;
13 % Define conversion range from 0 to 1
14 %conversion = h2; % Calculate conversion based on h2
15 %h2= (-alpha/2*p)*(1+eps*X);
16 %Dp= [.1, 10];
17
18 function h = exam1_Q2(w,y)
19 X= y(1);
20 p= y(2);
21
22 % parameter
23 %since Ct0 given and only A then it must be Ca0
24 Ca0= 2.0;
25 Fa0= 10.0 ;
26 k=10.0 ;
27 Dp= [.1, 10]
28 alpha= 1/Dp^2;
29 eps= 3.0 ;
30
31
32 Ca=((Ca0*(1-X))*(1+eps*X))*p;
33 Cb=(Ca0*(1+X)/(1+eps*X))*p;
34 Cc=((Ca0*(3+3*X))/(1+eps*X))*p;
35
36
37 % rate eq
38 ra= -k*Ca - k*Cb*(Cc^3);
39
40 %eqns
41 h1= -ra/Fa0 ;
42 h2= (-alpha/2*p)*(1+eps*X);
43
44 h= [h1, h2]';
45
46 end
```

Exam1_Q2_sim__1.m

```
1  clc;
2  clear all
3
4  % weight of particle we care about
5  wspan = [ 0 , 10];
6  % initial conditions for w and y
7  y0= [ 0 1];
8  % runs differential solver Rungekutta
9  [w,y] = ode45(@exam1_Q2,wspan,y0) ;
10
11 % Plot the results
12 figure(1);
13 plot(w, y(:, 1));
14 xlabel('Weight');
15 ylabel('conversion');
```